

Petroleum Subsidy Reform According to HANK: Case of Indonesia

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Outline

- 1 Introduction
- 2 Economic Model
- 3 Calibration
- 4 Results
- 5 Conclusion

Section 1:

Introduction

Background

- Supply shocks, particularly energy price shocks, have been a defining feature of the global economic cycle over the past decade.
- Monetary policy is limited, the attention has been directed toward the role of fiscal policy, such as energy subsidies, as a stabilisation instrument.
- However, studies on the role of fiscal policy as energy price shock absorber remain limited, especially for the case of developing countries.
- This paper studies economic implications of petroleum subsidy policy in **Indonesia**.

Institutional Context

- First introduced in the late 1970s, petroleum subsidy has remained a central fiscal instrument in Indonesia.
- **Mechanism:** The subsidy policy enables all consumers to purchase certain petroleum products at subsidised price, which is below the market price. The government covers the price differentials.
- **Subsidised Price Determination:** The subsidised petroleum prices are relatively rigid and do not automatically adjust to global price fluctuations (i.e., *ad hoc* price adjustment).
- **Consequence:** Large fiscal burden. [▶ See the Figure](#)

Research Objective

Evaluate the capacity of current petroleum subsidies in Indonesia (**ad hoc price adjustment** or baseline policy regime hereafter) by comparing it to alternative fiscal policy regimes along **three dimensions**.

- The long run macroeconomic, fiscal, and equity-related implications.
- The transitional consequences of subsidy reform.
- Short run macroeconomic stabilisation and fiscal sustainability in response to shocks.

Literature Review

- ① Related to literature about the role of fiscal policy in presence of energy shocks. However, these studies primarily focus on the case of developed countries.
 - Pieroni (2023) and Glocker and Wegmüller (2024) calibrated their model using European countries while Auclert et al. (2023) uses the US economy and Erceg et al. (2024).
 - This paper focuses on developing country, Indonesia.
- ② Several works have examined the role of energy subsidies in developing countries, although still limited.
 - Ginn (2024) studies the petroleum subsidies in Malaysia and Plante (2014) in several developing countries.
 - This paper considers a broader set of policy regimes and focuses not only on long run implications but also on short run stabilisation dynamics and transitional effects of reform.

Literature Review (*cont.*)

③ Related to Indonesian case studies.

- Clements et al. (2003) and Yusuf et al. (2017) investigate the real and distributional impact of petroleum price liberalisation in Indonesia. Dartanto (2013) examines the implication of eliminating petroleum subsidies on fiscal balance.
- Extending the analysis by examining not only the long run outcomes but also the transitional effects of policy reforms and the short run stabilisation dynamics.
- This paper also considers a broader fiscal policy regimes.
- The first to employ a HANK model to analyse petroleum subsidies in the Indonesian context.

Section 2:

Economic Model

Environment

Model Description

This article uses one-asset **Heterogeneous Agent New Keynesian (HANK)** model, consisting of heterogeneous households, union (and recruiting firm), intermediate goods producers, final goods producers, government (fiscal policy), and central bank (monetary policy).

- 1 **Households.** Households are ex-post heterogeneous, exposed by idiosyncratic shocks, and buy government bonds but cannot fully insure themselves.
- 2 **Union and Recruiting Firm.** Union interacts with competitive recruiting firms determining labor supply. I impose nominal wage rigidity.
- 3 **Intermediate Goods Firms.** Intermediate firms use labor and petroleum as their inputs, set the price and supply the goods to final goods producers. Price rigidity exists.
- 4 **Final Goods Producers.** Firms are competitive and aggregating the intermediate goods.
- 5 **Government.** The government issues bonds, and sets tax rate, government consumption, transfers, and policy on petroleum subsidies.
- 6 **Central Bank.** CB sets the nominal interest rate according to Taylor rule.

Policy Settings

Policy Settings

- ➊ **Ad hoc Price Adjustment (Baseline).** The government adjusts the subsidised price of petroleum less frequently and still provides the subsidies in the long run.
- ➋ **No subsidies.** Government provides no subsidies so the prices are fluctuated according to the market price.
- ➌ **Automatic Price Adjustments.** The government will provide no subsidies in the long run but still manage to constrained the fluctuation of subsidised price caused by market price shocks.
- ➍ **(Mean-tested) Targeted Transfers.** Instead of providing price subsidies, the government gives cash transfers to households according to their income level.
- ➎ **Lump Sum Transfers.** The government provides (lump sum) universal cash transfers (i.e., everyone gets the same amount of transfers).

Households

Households are ex-post heterogeneous maximizing the life-time utility

$$V(e_t, b_{t-1}) = \max_{c_t, b_t} \frac{c_t^{1-\gamma}}{1-\gamma} - \frac{N_t^{1+\psi}}{1+\psi} + \beta \mathbb{E}_t V(e_{t+1}, b_t)$$

$$\text{s.t. } b_t + (1 + \tau^c)c_t = (1 + r_t)b_{t-1} + (1 - \tau_t^N)w_t e_t N_t + d_t + tr_t, \\ b_t \geq 0,$$

$$\log(e_{t+1}) = \rho_e \log(e_t) + \varepsilon_{e,t+1}, \quad \varepsilon_e \sim N(0, \sigma_e), \quad \mathbb{E}(e_t) = 1$$

c_t and p_t are composite real consumption and price level consisting of non-petroleum c_f and petroleum consumptions c_e , defined by CES aggregating technology

$$c_t = \left(\alpha^{\frac{1}{\sigma^h}} c_{f,t}^{\frac{\sigma^h}{\sigma^h-1}} + (1-\alpha)^{\frac{1}{\sigma^h}} c_{e,t}^{\frac{\sigma^h}{\sigma^h-1}} \right)^{\frac{\sigma^h-1}{\sigma^h}} \\ p_t = \left(\alpha p_{f,t}^{1-\sigma^h} + (1-\alpha) \bar{p}_{e,t}^{1-\sigma^h} \right)^{\frac{1}{1-\sigma^h}}$$

where σ denotes elasticity of substitution and α represents the weighted.

Labour Market Structure and Wage Setting

There is a continuum number of union normalizes from 0 to 1 aiming at maximize the overall households' economic benefit by choosing labor supply and nominal wage level

$$\max_{\{W_t(u), N_t(u)\}} \mathbb{E}_t \sum_{s=0}^{\infty} \beta^s \left[\int_0^1 \frac{W_{t+s}(u)}{p_{t+s}} - \frac{v'(N_{t+s}(u))}{u'(C_{t+s})} N_{t+s}(u) - \frac{\psi^W}{2} \left(\frac{W_{t+s}(u)}{W_{t+s-1}(u)} - 1 \right)^2 N_t du \right]$$

$$\text{s.t. } N_t(u) = \left(\frac{W_t(u)}{W_t} \right)^{-\theta^W} N_t$$

By symmetric equilibrium, we can define the following wage NKPC:

$$\psi^W \pi_t^W (1 + \pi_t^W) = \kappa^W \left(\frac{N_t^\psi}{C_t^{-\gamma}} - \mu^W w_t \right) + \beta \mathbb{E}_t \left[\pi_{t+1}^W (1 + \pi_{t+1}^W) \frac{N_{t+1}}{N_t} \right]$$

where $C_t = \int c_{h,t} d\Gamma_t$ is aggregate consumption.

Final Good Producers

There is a continuum competitive final good producers aggregating intermediate goods, with the following optimization problem:

$$\begin{aligned} \max_{Y_t, Y_t(f)} \quad & p_t Y_t - \int_0^1 p_t(f) Y_t(f) d\mathcal{F}_t \\ \text{s.t.} \quad & Y_t = \left(\int_0^1 Y_t(f)^{\frac{\theta^F - 1}{\theta^F}} d\mathcal{F}_t \right)^{\frac{\theta^F}{\theta^F - 1}} \end{aligned}$$

Given this problem, we can obtain demand for intermediate goods as follows

$$Y_t(f) = \left(\frac{P_t(f)}{P_t} \right)^{-\theta^F} Y_t$$

Intermediate Good Producers

The production function is defined by

$$Y_t(f) = A_t N_t^{\alpha^F} O_t^{(1-\alpha^F)}$$

Intermediate good producers maximize life-time profit by choosing inputs and price.

$$\max_{N_t(f), O_t(f), p_t(f)} \mathbb{E}_t \sum_{s=0}^{\infty} \Lambda_{t,t+s} p_{t+s} \left[\frac{p_{t+s}(f) Y_{t+s}(f)}{p_{t+s}} - w_{t+s} N_{t+s}(f) - \frac{p_{e,t+s}}{p_{t+s}} O_{t+s}(f) - \frac{\psi^F}{2} \left(\frac{p_{t+s}(f)}{p_{t+s-1}(f)} - 1 \right)^2 Y_{t+s} \right]$$

Where Λ is stochastic discount factor.

Intermediate Good Producers (cont.)

Recall that in Lagrangian form, term λ can be interpreted as marginal cost $mc_t(f)$. Therefore, we can obtain the input demand as

$$N_t(f) = \alpha_F \left(\frac{mc_t(f)}{w_t} \right) Y_t(f)$$

$$O_t(f) = (1 - \alpha_F) \left(\frac{p_t mc_t(f)}{p_{e,t}} \right) Y_t(f)$$

Lastly, by imposing symmetric equilibrium, we can define the following NKPC

$$\pi_t^F (1 + \pi_t^F) = \kappa^F (mc_t - \mu^F \frac{p_{f,t}}{p_t}) + \Lambda_t \mathbb{E}_t \left[\pi_{t+1}^F (1 + \pi_{t+1}^F) (1 + \pi_{t+1}) \frac{Y_{t+1}}{Y_t} \right]$$

Central Bank

Central bank follows Taylor-type rule by choosing nominal interest rate

$$i_t = r_{ss} + \phi_{\pi} \pi_t$$

Fisher equation links nominal to real interest rate

$$1 + r_t = \frac{1 + i_t}{1 + \pi_{t+1}}$$

The Government

- The government budget constraint in nominal terms is defined as

$$\tau_t^N w_t N_t + \tau_t^c c_t + b_t^g = (1 + r_t) b_{t-1}^g + G_t + tr_t + s_t$$

where $\tau_t^N w_t N_t + \tau_t^c c_t$ is total tax revenue, b_t^g is total bonds issued, b_{t-1}^g is matured bonds, G_t is government consumption, tr_t is transfers, and s_t is spending on petroleum subsidies.

- Total government spending on petroleum subsidies is thus given by

$$s_t = \frac{\bar{p}_{e,t} - p_{e,t}}{p_t} \left(\int c_{e,t} d\Gamma_t \right)$$

where $\bar{p}_{e,t}$ and $p_{e,t}$ be subsidised and non-subsidised petroleum prices such that $\bar{p}_{e,t} \leq p_{e,t}$, $c_{e,h,t}$ denotes petroleum consumption by household h , and $\mathcal{D}$ is the cumulative distribution function of the households.

The Government (cont.)

- To ensure that the equilibrium existence of government budget constraint, we assume that the fiscal policy is active through adjusting their income tax rate according to the following rule:

$$\tau_t^N = \tau_{ss}^N + \iota_b(b_{t-1}^g - b_{ss}^g)$$

where τ_{ss}^N and b_{ss}^g are steady state level of income tax rate and real bonds. $\iota_b \in (0, 1)$ is sensitivity coefficient.

Oil Market

- Indonesia is a price taker in oil market as it has become net-oil importing country in 2004.
- Given this fact, we consider that the market price of oil is exogenous following AR(1) process

$$\log(p_{e,t}) = \rho \log(p_{e,t-1}) + \epsilon_t, \quad \epsilon_t \sim N(0, \sigma_\epsilon^2)$$

where $\rho \in (0, 1)$ and $\mathbb{E}(p_e) = 1$.

Price Control

General Rule

Under price control regime, the government set the subsidised price of petroleum according to the following general rule:

$$\bar{p}_{e,t} = \omega \bar{p}_{e,t-1} + (1 - \omega) \varphi p_{e,t} \quad (1)$$

where $\omega \in [0, 1)$ regulates the persistence of the changes of subsidised price and $\varphi \in (0, 1]$ ensures that in the long run, the government is still providing subsidies. Additionally, $tr_t = 0$ under this regime.

Regimes

- 1 **Baseline:** $\omega \in (0, 1)$ and $\varphi \in (0, 1)$. For our baseline analysis to mimic Indonesian economy, we set $\omega = 0.9$ and $\varphi = 0.6$
- 2 **No Subsidies:** $\omega = 0$ and $\varphi = 1$
- 3 **Automatic Price Adjustments:** $\omega \in (0, 1)$ and $\varphi = 1$.

Transfers

General Rule

Under this regime, the government provides no subsidies, $\bar{p}_{e,t} = p_{e,t}$, but gives cash transfers contingent to their pre-transfer income level ($y_{h,t}$) according to this rule:

$$tr_{h,t} = \left(\frac{\varphi^{tr}}{(1 + \kappa^{tr} \exp(y_{h,t} - \bar{y}))} \right) \frac{p_{e,t}}{p_t} \int c_{e,t} d\Gamma_t \quad (2)$$

where $\varphi^{tr} \in (0, 1)$, $\kappa^{tr} \geq 0$, and $\bar{y}$ is the pre-transfer income cutoff. More specifically, φ^{tr} and κ^{tr} respectively regulate the maximum transfers derived the portion of aggregate petroleum consumption and the slope of the amount of transfer across income level.

Regimes

- 1 **Mean-tested Transfers:** $\varphi^{tr} \in (0, 1)$ and $\kappa^{tr} > 0$.
- 2 **Lump Sum Transfer:** $\varphi^{tr} \in (0, 1)$ and $\kappa^{tr} = 0$.

Market Clearing Condition

Aggregate Consumption:

$$C_t = \int g_{c,t}(b, e) d\Gamma_t$$

Bond market:

$$b_t^g = \int g_{b,t}(b, e) d\Gamma_t$$

Aggregate Resource Constraint

$$Y_t = C_{f,t} + G_t + \Phi_t^p + \Phi_t^w$$

Where Φ^p and Φ^w are respectively price and wage adjustment costs (a la Rotemberg).

Section 4:

Calibration

Model Parameters

Parameters in this model are calibrated to closely mimicking distributional and petroleum related variables of Indonesian economy: **(a) Wealth Gini coefficient; (b) Income Gini Coefficient; (c) Subsidy Expenditure to Government Consumption Ratio; and (d) Share of petroleum consumption.**

Model Parameters

Table: Model Parameters

Parameters	Description	Value	Source
Households			
β	Discount factors	0.938	Auclert et al. (2021)
α	Distribution parameter of non-petroleum consumption	0.965	To match petrol consumption share
σ^h	Elasticity of Substitution	0.5	Gagliardone and Gertler (2023)
ψ	Frisch elasticity of labor supply	1	Auclert et al. (2021)
γ	Coefficient of relative risk averse	2	Pieroni (2023); Auclert et al. (2021)
τ^c	Consumption tax	0.11	VAT rate in Indonesia
ρ_e	Idiosyncratic shocks persistence	0.96	To match Indonesian Gini
σ_e	Innovation of idiosyncratic shocks	0.2	To match Indonesian Gini
n_e	Number of productivity states	5	Author
Union and Firms			
θ^W	Constant elasticity of substitution of labor	10	Pieroni (2023)
Ψ^W	Cost of wage adjustments	75	Author
θ^F	Constant elasticity of substitution of input	10	Pieroni (2023)
Ψ^F	Cost of price adjustments	75	Author

Model Parameters (cont.)

Table: Model Parameters (cont.)

Parameters	Description	Value	Source
Policy			
φ^{tr}	Intensity of transfers (mean-tested)	0.705	Target: $TR = 0.0198$
	Intensity of transfers (lump sum)	0.613	Target: $TR = 0.0198$
κ^{tr}	Slope of mean-tested transfers	10	Author
φ_{π}	Policy rate responses to inflation	1.25	Passive Monetary Policy
Petroleum Price			
ρ_e	Petroleum price persistence	0.86	Author
σ_{ϵ}	Innovation of petroleum price	0.48	Author

Benchmark Model Performance

Table: Model Performance

	Model	Data
Gini Coefficient (Assets)	0.7326	0.7454
Gini Coefficient (Income)	0.3449	0.3674
Subsidies to Government Consumption	0.0765	0.0768
Share of Petroleum Consumption	0.0447	0.0443

Notes: The variables generated by the model is steady state value under baseline regime (price subsidies) while those generated by the data is the mean over certain periods. For Asset Gini Coefficient, the data is obtained from World Inequality Data (WID) from 2000-2023. Income Gini Coefficient is obtained from World Income Inequality Database (WIID). Ratio of subsidies to government consumption is collected from CEIC and the Audit Board of Indonesia throughout 2010-2023. Lastly, share of petroleum consumption obtained from Living Cost Survey 2022.

Section 5:

Results

Impacts of Long Run Portion of Subsidies φ

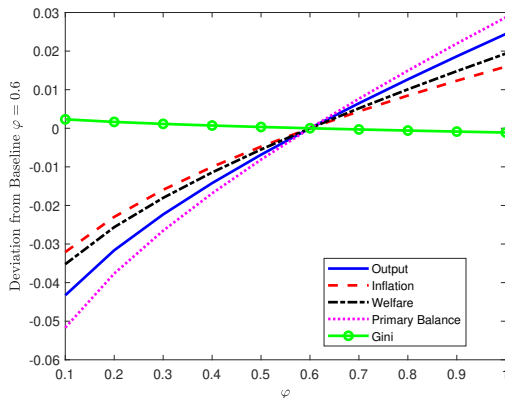


Figure: Impacts of φ on Macroeconomy, Primary Balance, Distribution, and Welfare

Notes: This figure exhibits the percentage difference between baseline portion of subsidies with $\varphi = 0.6$ to other portions. Higher φ indicates lower portions of subsidies paid by the government. In this Figure, the government does not provide transfers, $TR = 0$.

Cross-regime Comparisons: Price Control vs. Transfers

Table: Steady State Comparison

Variables	No Subsidies	Mean-tested Transfers	Lump Sum Transfers
Welfare	0.0194	0.0264	0.0251
Y	0.0244	0.0130	0.0149
p	0.0160	0.0160	0.0160
p_f	0.0000	0.0000	0.0000
c	0.0230	-0.0005	0.0028
c_f	0.0312	0.0075	0.0107
c_e	-0.2013	-0.2196	-0.2171
Primary Balance	0.0287	0.2101	0.1358
B^g	0.0452	0.0444	0.0438
τ^N	0.0000	0.1756	0.1540
Policy Rate	0.0000	0.1771	0.1056
N	-0.0085	-0.0196	-0.0178
O	-0.1916	-0.2005	-0.1991
w	0.0333	0.0333	0.0333
Profit	0.0083	-0.0029	-0.0011

Notes: Numbers in this table are representing the percentage difference from the baseline scenario. The steady state results for no subsidies and automatic price adjustment regimes are identical.

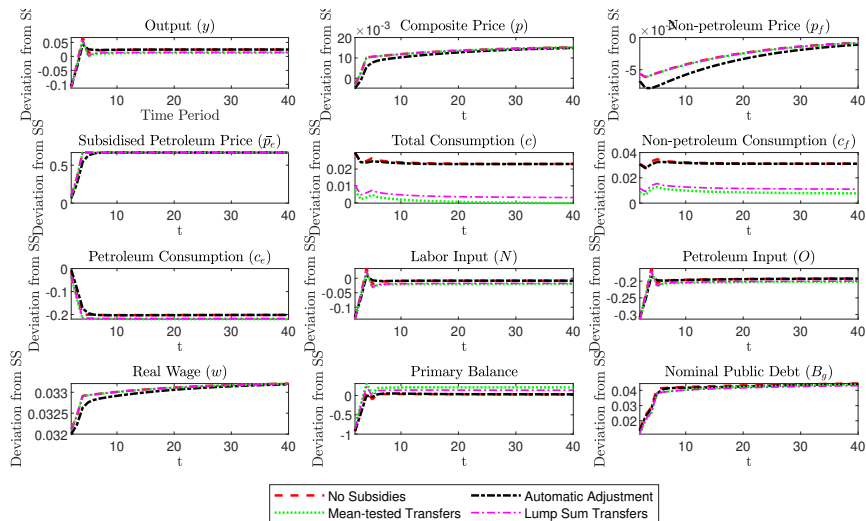
Distributive Aspects Comparison

Table: Distribution Coefficients

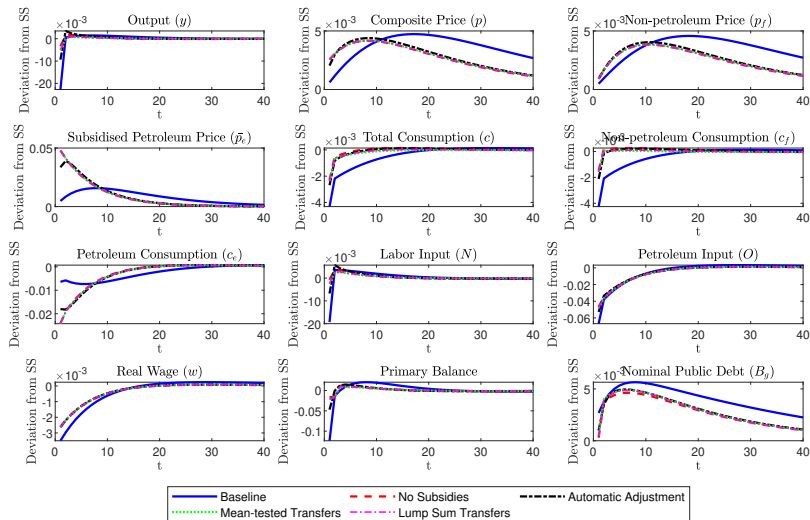
Variables	Baseline	No Subsidies	Mean-tested Transfers	Lump Transfers	Sum
Assets Gini Coefficient	0.7347	0.7338	0.7333	0.7346	
Assets Share of Top 10 Percent	0.5383	0.5373	0.5411	0.5368	
Assets Share of Bottom 50 Percent	0.0148	0.0151	0.0160	0.0144	
Assets Share Ratio	36.3483	35.4785	33.7794	37.2009	
Income Gini Coefficient	0.3451	0.3457	0.3396	0.3407	
Income Share of Top 10 Percent	0.2573	0.2575	0.2554	0.2559	
Income Share of Bottom 50 Percent	0.2682	0.2679	0.2717	0.2711	
Income Share Ratio	0.9592	0.9615	0.9403	0.9440	

Notes: The steady state results from no subsidies and automatic price adjustment regime are identical.

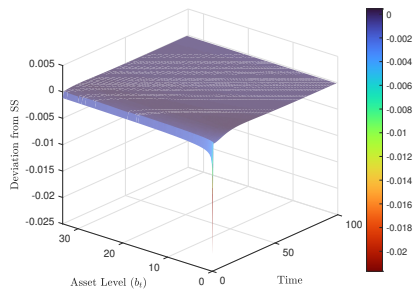
Transitional Dynamics of Policy Reform



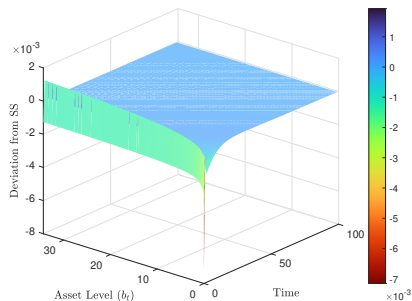
Stabilisation Outcomes



Wealth-specific Consumption Responses

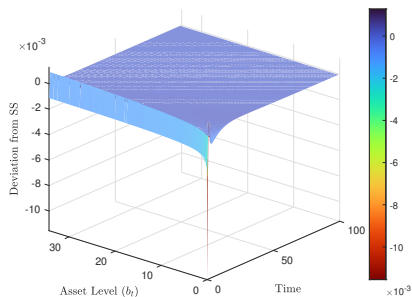


Baseline Regime

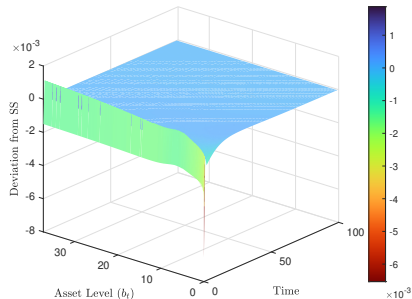


No Subsidies

Wealth-specific Consumption Responses (cont.)



Automatic Price Adjustment



Targeted Transfers

Wealth-specific Consumption Responses (cont.)

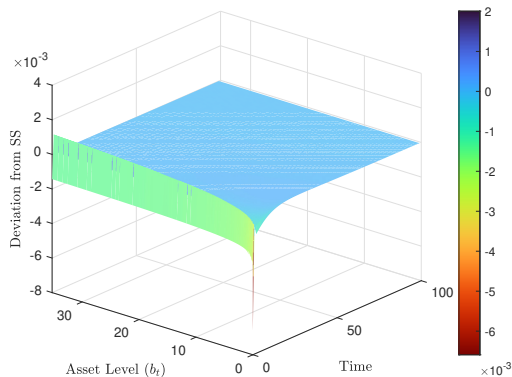


Figure: Lump Sum Transfers

Section 6:

Conclusion

Conclusion

- 1 Eliminating petroleum subsidies increases output, improves welfare, and promotes a more equal distribution of income and wealth in the long run.
- 2 Transfer-based policies, especially targeted transfers, can further improve welfare and equality in the long run, although with a slightly lower output than no subsidies regime.
- 3 However, transitioning away from the baseline entails short-run economic hardships, posing challenges for policy reform.
- 4 In the short run, the baseline regime is counterproductive: while it mitigates initial price surges, it leads to larger output losses, more persistent price increases, and greater fiscal deterioration in response to oil price shocks.
- 5 The baseline regime performs poorly to protect the poor from oil price shocks.

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Appendix 1:

Figures and Tables

Petroleum Subsidies 2010-2023

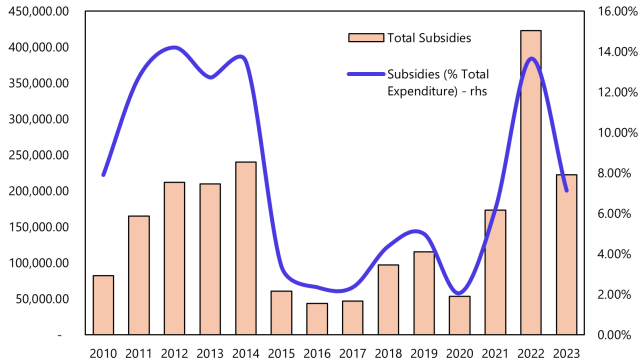
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Figure: Government Expenditure on Petroleum Subsidies in Indonesia, 2010-2023

Source: The Audit Board of the Republic of Indonesia.